

A Cybernetics Approach to Ethical AI Design

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Cybernetics and artificial intelligence share a long, intertwined history. This module explores the relationship between cybernetics and AI ethics, and looks at how cybernetics can be leveraged to reframe how we think about and how we undertake ethical AI design. The module is divided into three sections, beginning with an introduction to cybernetics. Following that, we explore different ways of thinking about AI ethics, before concluding by bringing the two concepts together to understand a new approach to ethical AI design.

Lecture Transcript

0:09 Hello everyone, I'm really excited to be contributing a module to AI Ethics: Global Perspectives. I'm going to use this module to explore relationships between cybernetics and AI ethics. And I'm going to do this in three parts. I'm going to start with an introduction to cybernetics. Then I'm going to give a little overview of different ways of thinking about and articulating AI ethics. And then I'm going to bring the two together to explore how cybernetics can help reframe how we think and undertake ethical AI design.

0:52 I'm going to start as we always do in Australia, and here at the School of Cybernetics with an acknowledgement of country and acknowledge that I'm speaking to you from Ngunnawal and Ngambri country in Canberra. I'd like to pay my respects to Aboriginal and Torres Strait Islanders, elders past and present, and to acknowledge that the land that I'm standing on has never been ceded. I've chosen this image for you today to articulate my acknowledgment of country, because it reflects some of the themes that we'll explore during this module. You're looking at an image of the Brindabellas, a mountain range just outside Canberra. When you look at it, you can see a range of things emerging in that image. You can see evidence of human engineering, human intervention. You can see that there's regeneration underway across the landscape. You can see evidence of technical systems, electricity infrastructure. You can see fire trails. You can also see an environmental ecosystem. And what is significant about this image is that it is also an image of kind of a past, a present and looking towards the future, because that particular area outside Canberra was completely devastated by the 2003 bushfires here just outside the city. And so what you're looking at is a story of regeneration, and reengagement with a landscape with a view to building a National Arboretum for generations to come. In this image are themes like human, ecological and technological systems intertwining in relation to each other. You have a story of the past and the present, and a future that unfolds before it. And you also have a sense of perspective - you can see that the photographer in this image is seeking out something particular. They have a vantage point, they bring a sense of something that they value, and try to capture. And what is it that they try to capture? They're trying to capture the light falling across the range. And so those concepts, the human, the ecological and the technical, intertwine a sense of perspective, a sense of past, present, and future are things that we're going to explore throughout this module.

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3:18 So let's start with part one. And look at the origin story for cybernetics. Cybernetics was founded by an MIT professor, Norbert Wiener, who was exploring self-guided missile systems and other forms of communication and control during World War Two. Cybernetics was a transdisciplinary endeavor. It brought together a range of disciplines dedicated to discussing relationships between humans and machines, and the environments that we sat within, understanding how these things related to each other, intersected with each other, provided information and received information in return. These kinds of questions were really pertinent in the context of World War Two, because in World War Two, we saw the development of incredibly advanced forms of weaponry. World War Two was where we saw the development of self guided missile systems for example, an example of a self learning system, able to gauge something about the environment it sat within and make a seemingly autonomous decision in response. Cybernetics influenced developments in anthropology, neuroscience, systems engineering, mechanical engineering, and a range of other theories and advancements in disciplines across the 1950s and 60s. It also included voices in the development of artificial intelligence. Some of those early contributors to discussions around cybernetics were present at that first research workshop in the mid 1950s at Dartmouth in the United States, contributing to the development of a new agenda for computing, called artificial intelligence. Cybernetics was inherently messy, it recognized that the systems that we develop, were always in dialogue with, responding to, influencing and being influenced by these dynamic systems that they sat within and intersected with. That's something that we've potentially forgotten over the past 75 years, developments in Computer Science and Artificial Intelligence. And so, it is a really pertinent moment to be exploring what it is cybernetics can tell us about AI today.

5:50 The founder of the School of Cybernetics, Distinguished Professor Genevieve Bell, has pioneered new cybernetics and attempted to reclaim the human and the ecological systems, alongside the technological systems that they interact with and are shaping them. We stop, we cease to center technological systems as our focus and instead think of them in relation to these broader contexts, these broader forces shaping our lives. Something that's also really crucial to new cybernetics is that we invoke both remembering, learning, stories of the past, understanding what history can tell us about some of the challenges that we're encountering now. It also invokes a sense of relearning, re-framing the way that we approach systems today, and reimagining, thinking how things could look differently in the future. In the 21st century, when Artificial intelligence has ceased to be something abstract, or mystical, the stuff of science fiction. Today, artificial intelligence, autonomous systems are part of how we go about our daily lives, how we work, they shape activities and processes across sectors. As I proceed through this module, I'm going to give you some things to read, watch and listen to, and some questions to ask yourself as you go. So if your first recommendation, at the end of part one, I'd like you to go away and read Distinguished Professor Genevieve Bell's essay for the *Griffith Review*, titled "Touching the Future". You'll see a link to it on the webpage accompanying this module. And as you read it, and as you reflect on this module, I'd like you to reflect on this question. New cybernetics brings together stories from the past, to understand the present that we find ourselves in, and to imagine a different future. It looks at relationships between humans and AI systems. So what stories and moments from your own history or from the histories that you've read into but that you're familiar with - what stories and moments does this make you think?

8:19 In part two, we're going to unpack ethical AI a little bit. But before we do, we need to pause for a moment and fix on a definition of artificial intelligence for the purposes of this section. There are many definitions of artificial intelligence. A definition that we often invoke here at the School of Cybernetics is a way of describing artificial intelligence that comes from the AI Now Institute in New York. And they simply refer to artificial

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intelligence as a constellation of technologies, processes, tools, techniques, that together bring about an autonomous, or a semi autonomous, way of approaching a process. So that constellation of technologies idea is a really cybernetic one, that there are a range of processes, roles, actors, technologies involved in building one system. It's not one homogenous thing.

9:25 In 2020, Carly Kind, who's the Director of the Ada Lovelace Institute in the United Kingdom, described three waves of ethical AI. In Wave one, she talked about philosophers and principles really driving the way we thought about ethical AI, are trying to lay the groundwork of principles that could shape the design of AI. In Wave two, we primarily became focused on technical interventions. How is it that we could rectify or address issues like bias in the outputs of automated systems? And then in Wave three, the wave that we're riding at the moment, she characterized this as a wave centering justice in AI, a wave that characterizes ethical AI in terms of dynamics like power, structural asymmetries that shape the contexts within which we're designing AI systems, and the injustices that occur. These three waves have pre-histories prior to the development of AI ethics as we understand it today. There have been a range of approaches to explore how it is that we design, interact and live with computing systems. These have included things like systems philosophy, value sensitive design, design ethics, and information ethics, which can be traced all the way back to Norbert Wiener that founder of cybernetics, who has been called by some the father of information ethics and information theory, as well.

11:07 So what is different about cybernetics? How does it reframe ethical AI? Typically, these waves of ethical AI that I referred to still center artificial intelligence, and try to determine either how it is that we produce an ethical system, or how it is that we assure ourselves we have acted ethically. Cybernetics asks us to step back. It takes a systems lens. And there's this beautiful quote, that comes from Norbert Wiener that you can see there on the slide, which articulates this way of seeing systems in cybernetics: "To describe an organism, we do not try to specify each molecule in it, and catalog it bit by bit, but rather try to answer certain questions about it which reveal its pattern". So what I'd like you to do, before we move into part three of this module, is note down a few things that I thought you could go away and watch. You'll also see them linked at the end of this activity. Two recommendations that I have for you. One is *Coded Bias*, a great documentary directed by Shalini Kantayya, and starring some really well known researchers, scholars and activists in the context of AI ethics, people like Joy Buolamwini, Cathy O'Neil, Safiya Noble, Meredith Broussard. And the second recommendation I have for you is a documentary from here in Australia called *Maralinga Tjarutja*. which is a documentary about the Maralinga people and their struggle to reclaim possession of and reckon with the history of atomic testing on their land. It's a beautiful story that brings together a range of kind of systems level effects of how it is that we deploy and design technology in human and ecological context. And as you reflect on part two of this module, a couple of questions for you: Which of those waves of AI ethics that I refer to were you already the most familiar with? And what do you think is missing from the discussion around AI ethics?

13:33 So now we're into Part Three, where we bring cybernetics and ethical AI together to explore how cybernetics can help reframe certain elements of ethical AI. So cybernetics helps us break down that concept of ethical AI, as a constellation of people and processes and infrastructure and responsibilities and motivations, that are all heart of ensuring that the AI that we live with is safe, responsibly designed, and responsive to injustice and harm, that it is, in fact, ethical in a dynamic, evolving sense. It allows you to recognize that ethical AI is not the role of one person or one team, but is in fact a dialogue, a relationship of cause and effect and then cause as a result of that effect, that are kind of always unfolding around both the systems that we develop, and the way that

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they're received in the world. Cybernetics gives us a sense of dynamism in acknowledging that the context that we operate within are ever changing. We recognize that what is perceived as ethical now, will not always be so because what is perceived as ethical is the product of our human values and assumptions. And those values and assumptions vary vastly by culture, context and personal circumstance. And so, rather than trying to achieve ethical AI as an outcome, as a horizon we're going to reach, instead we think of our responsibilities as designers, in terms of how it is that we know we're navigating in the right direction, how it is that we know when we've ceased to behave in an ethical fashion.

15:34 Cybernetics also gives us a language for talking about relationships, intersections, relationships between people and technologies, relationships between people, technologies and environments, and the effects of those relationships, what they're doing to each other. And it recenters our efforts as designers around not achieving perfect systems, but ensuring that the processes and practices that we use build in agility, responsiveness, resilience, building systems that are resilient in the face of the dynamic, changing contexts that they operate within.

16:22 So I'm going to leave you with three concepts from cybernetics that can help us reframe the way we think about AI ethics. The first is feedback, which is a really important concept that arose out of cybernetics. Our founder, Genevieve Bell, has this really lovely quote, that "The output of the system is always a future input". That whatever we do, has an effect. And that effect is informing our own practice. And actually, when you dwell on that concept, you see it play out in a number of the systems we interact with daily, in the feedback between recommendation systems, and people acting on those recommendations. And what those actions are telling the systems, trying to learn from and respond to them. We also start to understand the fragility or absence of really important feedback loops, that can tell us something about whether our AI system is still operating as we intended it to.

17:29 The second concept that I want to leave you with is causality as having a systemic character. It recognizes that in these dynamic, messy contexts we live within, that it's very difficult to draw a line from A to B. And that things that may seem simple to predict using data are actually, in practice, really complicated. The image that you're looking at on this slide is an image from the protests by UK students, when the outputs of final exams were automated in the first year of the pandemic. Rather than have students sit their exams, given the public health crisis that was unfolding, UK education authorities decided that, using data from past results, they could predict the likely outcomes of student exam results. And in fact, what occurred when those results were published, were realizations that those results were influenced by things like the location of schools at which those students had had previous results calculated, other classmates and their backgrounds, and those students own socioeconomic circumstances. These are all having an influence on the likelihood of the kind of performance that they would have in their final exams, as perceived by an AI system. But we actually know in reality is much more complicated than that. It is not as simple as we can take things about the past and use them to predict the future. That in fact, your success at university is not really just a reflection of your test results. That your test results are themselves the result of these interacting social and economic forces, and that the university itself has a number of factors and forces and systems that are going to shape your future success. It's never as simple as we can take data about the past and use it to prophesize about the future.

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19:39 The third concept that I want to leave you with from cybernetics is about perspectives. A really important transition from first order to second order cybernetics was the realization that we were not observing systems from somewhere else, somewhere objective and apart from the automated systems we were trying to engineer or the humans that we were trying to study. But instead that we were also part of the systems that we observed, that we brought a vantage point to that. And there's this great quote that you'll see on the slide from Heinz von Foerster, an incredibly well known physicist and mathematician, and an early cyberneticist, who said about perspective and stance: "I mentioned objectivity before, and I mention it here again as a popular device for avoiding responsibility...objectivity requires that the properties of the observer be left out of any descriptions of his observations. With the essence of observing, namely the processes of cognition". So what does this mean in the context of the kinds of AI systems being developed and applied today? Well, here's one example on the slide that you can see, which was a story documented about the failures of an AI system developed by Google to diagnose eye disease from images of the eye. When it was transposed from the lab to a real life environment, in that messy human context, a number of factors were influencing and affecting the performance of that system. Everything from technical infrastructure considerations like internet speeds, ability to access computers during diagnostic exams, geographic distances between clinics, and how people interpreted and responded to those results. All of these kinds of factors and cybernetics would be drawn on in designing an approach to deployment. It is not that we cease to deploy, but we develop processes and practices that help us to anticipate and as a result, build more resilient and sustainable systems that are more likely to succeed when introduced in these real world contexts.

22:07 So at the end of part three, where I brought cybernetics and AI ethics together, I'm giving you something to go away and listen to. And again, you'll find the link at the bottom of this module. This particular podcast is from 2017. It's from *The Guardian's* podcast "Chips with Everything", and it's a look back at the rise of British cybernetics, and the Ratio Club. This Ratio Club laid the foundations for the British cybernetics movement, and ultimately artificial intelligence. And so I wanted to ask you, looking forward to 2022 - What new discussions could emerge from cybernetics today and who needs to be part of those conversations?

22:55 The School of Cybernetics is based in Canberra, Australia. You're looking at a beautiful image of our Birch building. We are a transdisciplinary school - we bring together computer scientists, roboticists, educators, lawyers, journalists, theater makers, nuclear physicists, a range of people with different kinds of expertise, who share a common mission and a set of values. And that is using cybernetics to help us solve complex problems in the world. And in particular, to help us develop a new branch of engineering to manage AI systems safely, sustainably and responsibly. We have education programs spanning microcredentials all the way through to a masters and a PhD program. And we also undertake a range of research projects exploring everything from art and technology to new forms of responsible data practice, to reimagining the role of libraries as information users, archivists and providers in the 21st century. We are together dedicated to creating a sustainable, enduring set of practices and approaches to cultivate responsible, sustainable and ethical AI. Thank you for watching this module. I hope you enjoyed it. If you wanted to continue the conversation, you can check out the School of Cybernetics website. I look forward to talking further about the relationship between cybernetics and ethical AI.